

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Implemented and developed **NPC AI systems and gameplay mechanics** using **C++ and custom engine architecture**, delivering high-performance game code across **PC and console platforms** with profiling and debugging tools; reduced memory footprint by **18%** through optimized AI state management.
- Built and maintained **UI systems and general gameplay features** for AAA game titles using **source control (Git) and bug reporting software**; executed prioritized tasks in Agile team environments, collaborating with art, animation, and design disciplines to integrate **3D mathematics** and asset pipelines.
- Engineered **high-performance, well-documented gameplay code** with expertise in **Unreal Engine** systems including AI behavior trees, gameplay frameworks, and console optimization; achieved **60 FPS stability** across PlayStation, Xbox, and PC through iterative testing and deployment cycles.
- Deployed and tested features on **PC and consoles** using profiling and debugging tools; provided engineering support across interconnected disciplines through clear communication and problem-solving, reducing post-deployment issues by **22%** and supporting UAT alignment with design and art teams.
- Applied **3D mathematics, AI system design, and gameplay implementation** through shipped game jam projects and college portfolio work demonstrating mastery of **Unreal Engine gameplay and AI systems**; created reusable gameplay components achieving **40% code reuse** across multiple game modules.
- Leveraged **C++, Python, and C#** across game systems development, combined with **Agile development methodologies** and tool creation; delivered organized, well-documented solutions with strong problem-solving discipline, enabling teams to ship features on schedule and maintain code quality standards.

SKILLS

Game Development & Engines – Unreal Engine, custom engine architecture, gameplay systems, AI behavior trees, NPC AI, UI systems, console optimization.

Core Programming Languages – C++, C#, Python, 3D mathematics, game physics.

Game Development Tools & Workflows – Source control (Git), bug reporting software, profiling tools, debugging tools, PC and console deployment, testing and QA.

Gameplay & AI Systems – AI system design, AI state management, NPC behavior implementation, gameplay mechanics, game loops, asset integration.

Team & Project Methodologies – Agile development, cross-disciplinary collaboration, tool creation, problem-solving, technical documentation, code organization.

Additional Technical Skills – Memory optimization, performance profiling, multi-platform development, game architecture patterns, gameplay feature delivery.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin

02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India

01/2021

PROJECTS

AAA Game AI & Gameplay Systems Architecture (HCL Technologies)

- Architected and implemented **NPC AI behavior systems and gameplay mechanics** for a shipped AAA console title using **Unreal Engine** with advanced **3D mathematics** for pathfinding, collision detection, and dynamic state management.
- Delivered **high-performance, well-documented C++ code** achieving **60 FPS stability** across **PlayStation, Xbox, and PC platforms** through iterative profiling, debugging, and optimization; reduced draw calls by **32%** and memory overhead by **18%** using advanced engine profiling tools.
- Collaborated with art, animation, and design disciplines to integrate **3D assets, animation state machines, and gameplay events**; provided engineering support reducing design iteration cycles by **25%** and enabling seamless **UI system integration** across player-facing features.
- Managed source control workflows, bug reporting, and feature deployment across **PC and console test environments** using Agile sprint ceremonies; shipped **12+ gameplay features** on schedule with zero post-launch critical issues through comprehensive testing and UAT alignment.
- Implemented **tool creation scripts in C++ and Python** for automated gameplay validation, asset pipeline integration, and debugging; reduced manual QA time by **40%** and enabled rapid iteration on AI and gameplay systems across distributed team members.

Real-Time Game Performance Profiling & Optimization Pipeline (HCL Technologies)

- Built a **profiling and debugging pipeline** integrating **Unreal Engine's native profiling tools** with custom **C++ analysis modules** to track memory, CPU, and GPU performance across **PC and console builds** in real time.
- Reduced memory leaks and performance bottlenecks by **28%** through systematic debugging and optimization, enabling stable **60 FPS gameplay** on all target platforms and improving player experience in shipped titles.
- Delivered clear technical documentation and shared performance insights across engineering, art, and design teams, supporting root-cause analysis on **12+ high-impact gameplay and AI system bottlenecks** with actionable optimization recommendations.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)